A Project Report on

Computer Vision Based Touchless Device Control

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

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CERTIFICATE

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Abstract

The rapid advancement of digital technologies has significantly increased the reliance on traditional human–computer interaction methods such as keyboards, mice, and touchscreens. However, these input mechanisms require fine motor control and direct physical contact, making them unsuitable for individuals with motor impairments and potentially unsafe in shared environments such as public kiosks and healthcare systems. This project presents a computer vision-based touchless device control system that enables users to interact with digital systems using face and hand gestures, eliminating the need for physical input devices.

The proposed system adopts a hybrid interaction approach by integrating face-based and hand-based control mechanisms. Facial landmarks are used to track head movement for cursor control, while blink detection enables click actions and a mouth-based clutch mechanism prevents unintended cursor movement. In parallel, hand gesture recognition is implemented using landmark-based detection to support cursor movement, clicking, scrolling, and virtual keyboard input. A launcher-based architecture ensures efficient mode selection and prevents system conflicts.

The system is developed using Python with OpenCV, MediaPipe, NumPy, and PyAutoGUI, and operates entirely on edge devices without requiring cloud processing or pre-trained datasets. This ensures low latency, privacy preservation, and real-time performance. The proposed solution enhances accessibility for motor-impaired users and promotes hygienic interaction in public environments, aligning with Sustainable Development Goals focused on reduced inequalities and improved health and well-being.

Keywords: Human–Computer Interaction (HCI), Gesture Recognition, Face Tracking, Hand Gesture Control, MediaPipe, OpenCV, Real-Time Systems, Edge Computing, Accessibility, Motor-Impaired Assistance, Hygienic Interfaces, Cursor Control, Virtual Keyboard

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ABBREVIATIONS

HCI	*Human–Computer Interaction*
CNN	*Convolutional Neural Network*
PLML	*Partially-Latent Mixture of Linear Regressions*
GPU	*Graphics Processing Unit*
CPU	*Central Processing Unit*
HSV	*Hue, Saturation, Value*

Chapter 1

Introduction

1.1 Background

Human–Computer Interaction (HCI) has traditionally relied on physical input devices such as keyboards, mice, and touchscreens to facilitate communication between users and digital systems. These input mechanisms require direct physical contact and precise motor control, which can pose significant challenges for individuals with motor impairments. With the rapid expansion of digital systems across domains such as healthcare, education, public kiosks, and enterprise environments, the demand for more efficient and user-friendly interaction methods has increased significantly.

Conventional input devices are not always suitable for all users, particularly those with limited mobility or physical disabilities. Additionally, in shared environments, repeated physical contact with input devices raises hygiene concerns. These limitations highlight the need for alternative interaction approaches that are both accessible and safe for widespread use.

1.2 Motivation

One of the primary motivations for this project is to enhance accessibility for motor-impaired individuals. Many existing systems require fine motor control, making it difficult for such users to interact effectively with digital interfaces. By enabling control through facial movements and hand gestures, the system reduces dependency on physical devices and allows users to interact more naturally and independently. This aligns with **Sustainable Development Goal (SDG) 10: Reduced Inequalities**, which focuses on inclusive access to technology.

Another key motivation is the increasing need for hygienic interaction methods in public and shared environments. Devices such as kiosks, ATMs, and hospital terminals are used by

multiple users, increasing the risk of contamination. A touchless interaction system minimizes physical contact and promotes safer usage, aligning with **SDG 3: Good Health and Well-being**, which emphasizes reducing health risks and ensuring safer environments.

1.2.1 Need for Touchless Interaction Systems

The growing integration of digital systems into everyday life has made interaction methods a critical aspect of user experience. Traditional interaction mechanisms are not only limited by physical constraints but also fail to address modern requirements such as accessibility, hygiene, and convenience. Touchless interaction systems provide a promising alternative by leveraging computer vision techniques to interpret user gestures and facial movements in real time.

Such systems eliminate the need for physical contact and enable intuitive interaction without additional hardware. They are particularly useful in environments where hygiene is a priority and in scenarios where users may have limited physical capabilities. The development of efficient and reliable touchless systems can significantly improve usability and expand the reach of digital technologies.

1.3 Objectives

The primary objective of this project is to design and develop a real-time touchless device control system using computer vision techniques. The system aims to integrate both face-based and hand-based interaction methods into a unified framework to enhance usability and flexibility.

Specific objectives include enabling cursor movement through head tracking, implementing click actions using blink detection, and introducing a mouth-based clutch mechanism to improve control stability. Additionally, the system supports hand gesture-based interactions such as cursor navigation, clicking, scrolling, and virtual keyboard input. The project also focuses on developing a lightweight, edge-based solution that operates without reliance on cloud processing or specialized hardware.

1.4 Scope of the Project

The scope of this project includes the development of a real-time touchless interaction system that operates using standard webcam and software-based processing. The system is designed to function on general-purpose computing devices without requiring additional sensors or external hardware.

The project focuses on implementing core functionalities such as gesture detection, facial tracking, cursor control, and system interaction using computer vision techniques. While the system performs effectively under normal conditions, it may be influenced by environmental factors such as lighting variations and camera quality. Future enhancements may include improved robustness, adaptive calibration, and integration of advanced gesture recognition techniques.

Chapter 2

Literature Review

The development of touchless human–computer interaction systems has gained significant attention in recent years due to advancements in computer vision and the increasing demand for accessible and hygienic interaction methods. Various research efforts have explored gesture recognition, facial tracking, and multimodal interaction systems to replace traditional input devices. These studies form the foundation for the proposed system and highlight both the progress made and the limitations that still exist in this domain.

2.1 Literature Review of Existing Systems

The development of touchless human–computer interaction systems has gained significant attention due to advancements in computer vision and the increasing demand for accessible and hygienic interaction methods. Various studies have explored gesture recognition, facial tracking, and multimodal interaction systems to replace traditional input devices.

Drouard et al. [1] proposed a regression-based approach for head pose estimation using PLML, achieving accurate real-time predictions but lacking gesture-based interaction. Kodali et al. [2] developed a gesture-based virtual mouse system using OpenCV and MediaPipe; however, it is sensitive to environmental conditions and lacks hybrid capabilities. Patle et al. [3] introduced a multimodal system integrating gestures, voices, and gaze, but it requires calibration and is resource

intensive. Hema et al. [4] demonstrated a hand gesture system with high accuracy under optimal lighting, though performance degrades in low-light conditions.

Zhang et al. [5] presented MediaPipe Hands, enabling efficient real-time hand tracking using deep learning-based landmark detection. While highly accurate, it focuses only on tracking and not gesture interpretation. Ruiz et al. [6] proposed a deep learning-based head pose estimation method that provides accurate orientation prediction but does not integrate gesture-based interaction.

A summary of the literature is presented in Table 1. The analysis indicates that most existing systems focus on single interaction modality and lack a unified hybrid framework. Additionally, they are affected by environmental conditions or require significant computational resources. These limitations highlight the need for a lightweight, real-time hybrid system integrating both face and hand-based interaction, which is addressed in the proposed work.

Table 1 Summary of Literature Survey

Author & Year	Techniques / Method Used	Dataset Used	Key Results	Limitations
Drouard et al., 2016	Partially Latent Mixture of Linear Regressions (PLML), Face bounding box-based head pose estimation from single RGB image	Public head pose datasets (e.g., AFLW, BIWI)	Robust head pose estimation with improved regression accuracy; suitable for real-time orientation prediction	Focused only on head pose; no gesture integration; computational complexity; not optimized for lightweight multimodal HCI
Kodali et al., 2023	OpenCV, MediaPipe hand landmarks, Haar Cascade face detection, Pycaw audio control, Rule-based gesture classification	Custom real-time webcam data	Real-time gesture-based volume control and virtual mouse; integrated facial authentication	Sensitive to lighting and occlusion; no adaptive learning; limited multimodal capability

Patle et al., 2025	Multimodal system (Gesture + Voice + Gaze), MediaPipe hand tracking, Vosk/Whisper speech recognition, Webcam-based gaze tracking, Context-aware modality switching, Edge deployment	Lab-based case studies (Healthcare, Classroom, Industrial Control Room, VR environments)	Adaptive modality switching based on environment; real-time multimodal interaction; privacy-preserving on-device processing; improved hygienic interaction	Requires calibration (especially gaze); affected by lighting and ambient noise; hardware resource constraints; limited large-scale deployment
Hema et al., 2025	OpenCV + MediaPipe (21 hand landmarks), HSV color segmentation, Euclidean distance thresholding, Virtual mouse, Air canvas, PPT controller	50 participants performing 20 predefined gestures each	93% accuracy under optimal lighting 89% accuracy in low light; real-time responsiveness	Performance decreases in poor lighting; occlusion sensitivity; mainly static gesture recognition; no facial interaction integration
Zhang et al., 2020	MediaPipe Hands, deep neural network-based hand landmark detection, real-time tracking pipeline, lightweight on-device inference	Large-scale annotated hand datasets	Accurate real-time hand tracking with 21 landmarks; efficient edge-device performance	Focus limited to hand tracking; no gesture interpretation or multimodal integration
Ruiz et al., 2018	CNN-based head pose estimation without keypoints, regression.	AFLW2000 dataset	High-accuracy real-time head pose estimation without explicit landmark detection	Requires training data; no gesture-based interaction or hybrid system support

Chapter 3

Limitation of Existing System

Despite advancements in gesture-based and vision-based interaction systems, existing solutions face multiple challenges that hinder their effectiveness in real-world scenarios. Issues such as lack of hybrid integration, sensitivity to environmental conditions, and absence of control stabilization mechanisms reduce their usability and reliability. These limitations highlight the need for a more robust and unified approach to touchless interaction.

3.1 Research Gap

Although several touchless interaction systems have been developed using hand gesture recognition, facial tracking, and multimodal interfaces, no single system provides a unified, lightweight, and accessible solution tailored for real-world deployment. Existing systems are either limited to a single interaction modality, require complex calibration, or depend on high computational resources.

Many gesture-based systems focus only on hand tracking for virtual mouse control, while face-based systems primarily deal with head pose estimation without integrating gesture interaction. Multimodal systems, although more flexible, often introduce complexity through calibration requirements and hardware dependencies. As a result, current solutions fail to provide a consistent, scalable, and user-friendly touchless interaction experience.

3.1.1 Identified Gap

From the analysis of existing systems, it is evident that most solutions fail to deliver a robust, hybrid, and real-time touchless interaction system suitable for both accessibility and hygienic use. The common shortcomings include:

- Lack of hybrid integration combining both face-based and hand-based interaction in a unified framework.
- Sensitivity to environmental conditions such as lighting variations, background noise, and occlusion.
- High computational requirements, making them unsuitable for edge devices and real-time applications.
- Absence of control stabilization mechanisms, leading to unintended cursor movements and reduced usability.
- Dependence on calibration procedures, particularly in gaze-based or multimodal systems.
- Limited focus on accessibility, making interaction difficult for motor-impaired users.

3.1.2 Research Gap Summary

There is no existing system that effectively combines:

- Real-time face-based control (head movement, blink detection) with hand gesture-based interaction.
- Lightweight, edge-based processing without reliance on cloud computation or specialized hardware.
- Control stabilization techniques such as pause or clutch mechanisms to improve interaction accuracy.
- A unified and user-friendly framework designed specifically for accessibility and hygienic interaction.

This research gap highlights the need for the proposed system, which addresses these limitations by integrating face and hand gesture recognition into a hybrid, real-time, and efficient touchless interaction framework. The system is designed to operate on standard hardware while ensuring stability, usability, and accessibility for diverse users in both personal and public environments.

Table 2 Identified Limitations and Research Gap

Limitation in Existing Systems	Impact	Research Gap Identified
Sensitivity to lighting and environmental conditions	Reduced accuracy in real-world scenarios	Need for robust system adaptable to varying conditions
Single-modality interaction (only hand or face)	Limited flexibility and reliability	Requirement for hybrid face + hand interaction system
High computational requirements	Not suitable for edge devices or public deployment	Need for lightweight, real-time system
Lack of control stabilization	Unintended cursor movement, poor usability	Need for clutch/pause mechanisms
Calibration dependency (gaze/voice systems)	Difficult setup and poor user experience	Need for calibration-free interaction
Lack of accessibility-focused design	Difficult for motor-impaired users	Need for simplified and inclusive interaction model

Chapter 4

Problem Statement and Objectives

The increasing demand for accessible and hygienic interaction systems has exposed the limitations of traditional input methods and existing touchless solutions. Motor-impaired users face difficulties in using conventional devices, while touch-based interfaces in public environments pose hygiene risks.

4.1 Problem Statement

1. Existing human–computer interaction systems rely heavily on physical input devices such as keyboards and mice, which require precise motor control and continuous physical effort.
2. These traditional input methods are not accessible to motor-impaired individuals, limiting their ability to interact independently with digital systems.
3. Public kiosks and shared digital interfaces depend on touch-based interaction, increasing hygiene risks and the potential spread of infections.
4. Current touchless interaction systems are mostly single-modality, focusing either on hand gestures or facial tracking, and lack a unified hybrid framework.
5. Many existing systems are sensitive to environmental conditions such as lighting, occlusion, and background noise, reducing their reliability in real-world scenarios.
6. There is a lack of control stabilization mechanisms, leading to unintended cursor movements and poor usability, especially for users with limited motor control.

Problem Statement: There is a need for a real-time, hybrid, and lightweight touchless interaction system that enables accessible, stable, and hygienic device control using face and hand gestures.

4.2 Objectives

- To develop a real-time touchless device control system using computer vision techniques.
- To integrate both face-based interaction (head movement and blink detection) and hand gesture recognition into a unified hybrid framework.
- To enable cursor movement, click operations, scrolling, and virtual keyboard input without the use of physical input devices.
- To implement a mouth-based clutch mechanism and pause control to improve interaction stability and prevent unintended actions.
- To design a lightweight and efficient system that operates on edge devices without requiring cloud processing or high computational resources.
- To improve accessibility for motor-impaired users by enabling intuitive and minimal-effort interaction.
- To promote hygienic interaction in public and shared environments by eliminating physical contact with input devices

Chapter 5

Proposed System

The limitations identified in existing touchless interaction systems highlight the need for a more robust, efficient, and unified solution. Current approaches lack hybrid integration, control stability, and real-world adaptability, making them less effective for practical deployment. To address these challenges, the proposed system introduces a computer vision-based framework that combines face and hand gesture interaction into a single architecture.

5.1 Proposed System

The proposed system is a computer vision-based touchless device control framework that enables users to interact with a computer using face and hand gestures. The system is designed to operate in real time using a standard webcam and does not require any external hardware or cloud-based processing.

The core idea of the system is to integrate both face-based and hand-based interaction into unified hybrid architecture. This improves flexibility and robustness, as the system can adapt to different user conditions and interaction preferences. The system is lightweight, efficient, and suitable for deployment on edge devices, making it practical for both personal and public use.

5.2 System Architecture

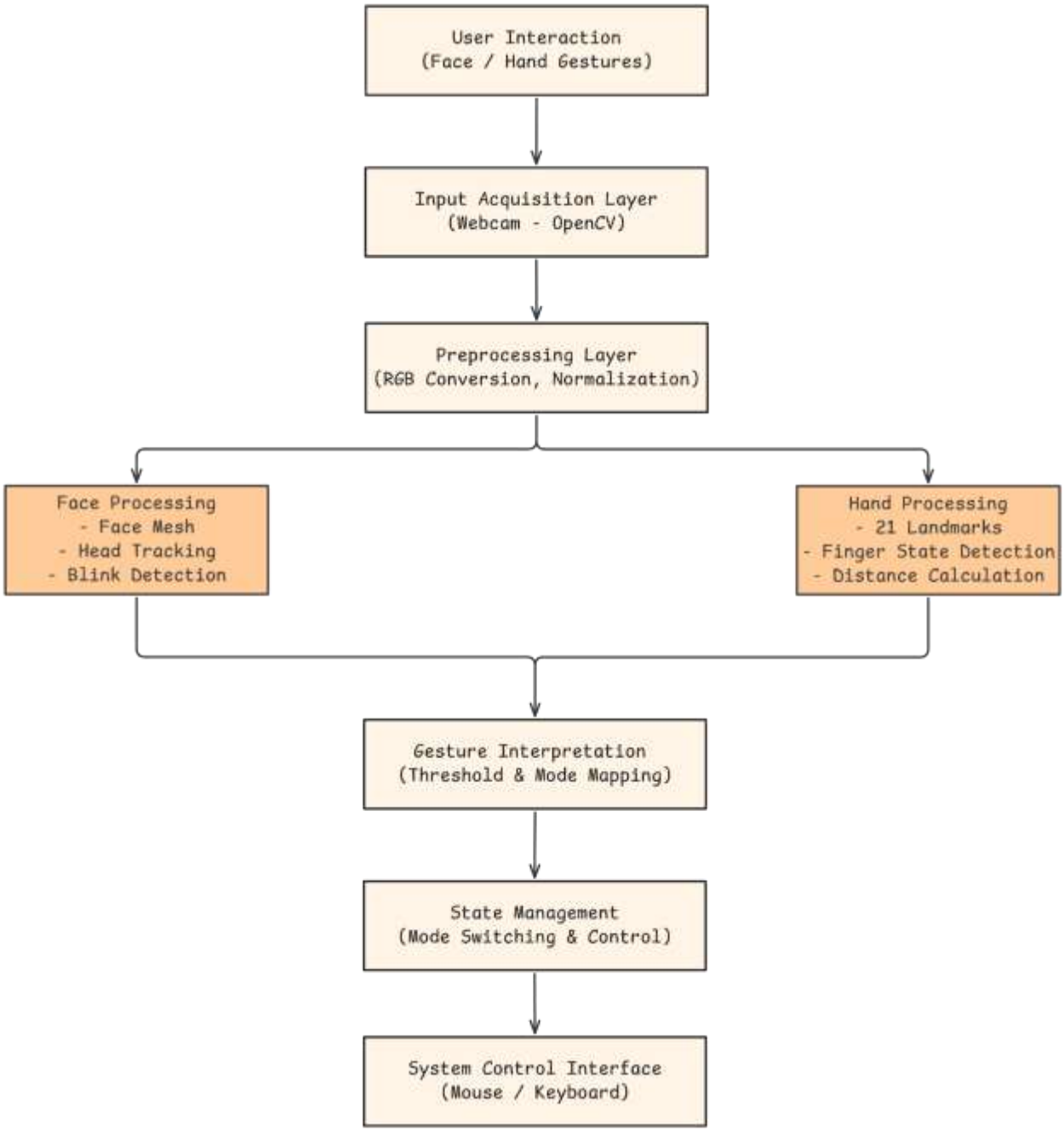


Figure 1 System Architecture

The system consists of the following layers:

- **Input Acquisition Layer:** Captures real-time video using OpenCV from the webcam.
- **Preprocessing Layer:** Performs basic operations such as frame normalization and format conversion to prepare the input for further processing.
- **Face Processing Module:** Uses MediaPipe Face Mesh to detect facial landmarks, track head movement, and detect eye blinks for click operations.

- **Hand Processing Module:** Uses MediaPipe Hands to detect 21 hand landmarks, determine finger states, and calculate distances for gesture recognition.
- **Gesture Interpretation Layer:** Applies threshold-based logic to interpret gestures and map them to specific actions.
- **State Management Layer:** Handles mode switching (mouse/keyboard), pause control, and system stability.
- **System Control Interface:** Executes actions such as cursor movement, clicking, scrolling, and keyboard input using PyAutoGUI.

The architecture of the proposed system follows a layered approach, as shown in the system design diagram. It begins with user interaction through face or hand gestures, which are captured using a webcam. The captured frames are processed and passed through different modules to interpret gestures and convert them into system-level actions.

5.3 Methodology

The system follows a real-time processing pipeline to convert user gestures into control actions. Initially, video frames are continuously captured from the webcam and preprocessed. These frames are then passed simultaneously to both face and hand processing modules.

In the face module, facial landmarks are extracted to track head movement, which is mapped to cursor position. Blink detection is used to perform click operations, while a mouth-based clutch mechanism temporarily pauses cursor movement to avoid unintended actions.

In the hand module, landmark positions are used to detect finger states and calculate distances between fingers. These features are used to identify gestures for cursor movement, clicking, scrolling, and keyboard interaction.

The outputs from both modules are processed by the gesture interpretation layer, where predefined thresholds and rules determine the final action. The state management layer ensures smooth operation by controlling modes, handling pauses and preventing conflicts between face and hand inputs. Finally, the system control interface executes the corresponding action on the device.

5.4 Design Details

The system is designed using modular architecture to ensure scalability and maintainability. Each module operates independently while interacting with others through a well-defined pipeline.

The face processing module focuses on stability and minimal movement by incorporating a clutch mechanism, which improves usability for motor-impaired users. The hand processing module is designed to support multiple gestures with minimal computational overhead using landmark-based detection.

The overall system avoids complex machine learning training pipelines by using pre-trained models from MediaPipe and rule-based gesture interpretation. This reduces computational requirements and enables real-time performance on standard hardware. The design ensures low latency, high responsiveness, and ease of deployment.

5.5 Advantages of Proposed System

- Hybrid integration of face and hand interaction for improved flexibility
- Real-time performance using lightweight computer vision techniques
- No requirement for external hardware or cloud processing
- Improved control stability using clutch and pause mechanisms
- Enhanced accessibility for motor-impaired users
- Hygienic, contact-free interaction suitable for public environments

Chapter 6

Experimental Setup

The experimental setup describes the environment, input data, and system configuration used to implement and evaluate the proposed touchless interaction system. It outlines the input source, performance evaluation criteria, and the software and hardware components required for real-time execution.

6.1 Input Data and System Input

The proposed system operates on real-time video input captured through a standard webcam. Unlike traditional machine learning-based systems, the proposed approach does not rely on pre-collected datasets or training data. Instead, it utilizes live video frames to detect and interpret user gestures dynamically.

The input consists of continuous image frames obtained from the webcam, which are processed using computer vision techniques. These frames are analyzed to detect facial landmarks and hand landmarks using the MediaPipe framework. The system extracts features such as head position, eye blink status, mouth opening, finger positions, and distances between landmarks. These features are then used to interpret gestures and map them to corresponding system actions.

6.2 Software Setup

Table 3 Software Setup

Component	Technology Used
Programming Language	Python
Computer Vision Library	OpenCV
Landmark Detection	MediaPipe (Face Mesh & Hands)
Numerical Computation	NumPy
System Control	PyAutoGUI
Development Environment	VS Code
Operating System	Windows / Linux

6.3 Hardware Setup

Table 4 Hardware Setup

Component	Specification
Processor	Intel i5 / Ryzen 5 or higher
RAM	8 GB or higher
Camera	Built-in / External Webcam
Storage	256 GB SSD or higher
GPU	Not required

6.4 Performance Evaluation Parameters

Since the proposed system is a real-time, rule-based interaction system, traditional machine learning metrics such as accuracy, precision, and recall are not directly applicable. Instead, the performance of the system is evaluated based on functional and usability parameters.

The key evaluation parameters include:

- **Response Time:** Measures the delay between gesture input and system action.

- **System Stability:** Evaluates the smoothness of cursor movement and consistency of gesture recognition.
- **Gesture Detection Reliability:** Assesses how accurately the system interprets intended gestures under varying conditions.
- **Robustness:** Determines system performance under different lighting conditions and backgrounds.
- **Usability:** Evaluates ease of interaction and effectiveness for motor-impaired users.

These parameters ensure that the system is evaluated in terms of practical performance and real-world usability rather than purely statistical measures.

Chapter 7

Results and Discussion

This chapter presents the results obtained from the implementation of the proposed touchless interaction system and analyzes its performance based on practical observations. The system is evaluated in terms of responsiveness, usability, and reliability under different operating conditions.

7.1 Observed Results

The proposed system was successfully implemented and tested in a real-time environment using a standard webcam. The system was able to accurately detect both facial and hand gestures and map them to corresponding system actions such as cursor movement, clicking, scrolling, and mode switching.

Face-based interaction enabled smooth cursor control using head movement, while blink detection was effectively used for clicking operations. The hand gesture module allowed users to perform multiple actions such as scrolling and gesture-based control using finger positioning. The integration of both modules provided a flexible and intuitive interaction experience.

The inclusion of a clutch and pause mechanism significantly improved control stability by preventing unintended cursor movements. Overall, the system demonstrated real-time responsiveness with minimal delay and performed reliably under normal lighting conditions.

7.2 Performance Analysis

The performance of the proposed system was evaluated based on practical observations during real-time execution. The system demonstrated near real-time responsiveness, with minimal delay between user gestures and system actions. Cursor movement using face tracking was smooth and stable, while blink detection provided reliable click operations.

The integration of the clutch and pause mechanism significantly improved control stability by reducing unintended cursor movements. Hand gesture recognition was effective under normal lighting conditions and allowed accurate execution of actions such as scrolling and mode switching.

Table 5 Performance Observations

Parameter	Observation
Response Time	Near real-time with minimal noticeable delay
System Stability	Smooth cursor movement with reduced jitter due to clutch mechanism
Gesture Recognition	Reliable under normal lighting conditions
Robustness	Performance slightly affected in low-light or occluded environments
Usability	Easy to use and suitable for motor-impaired users

7.3 Comparison with Existing Systems

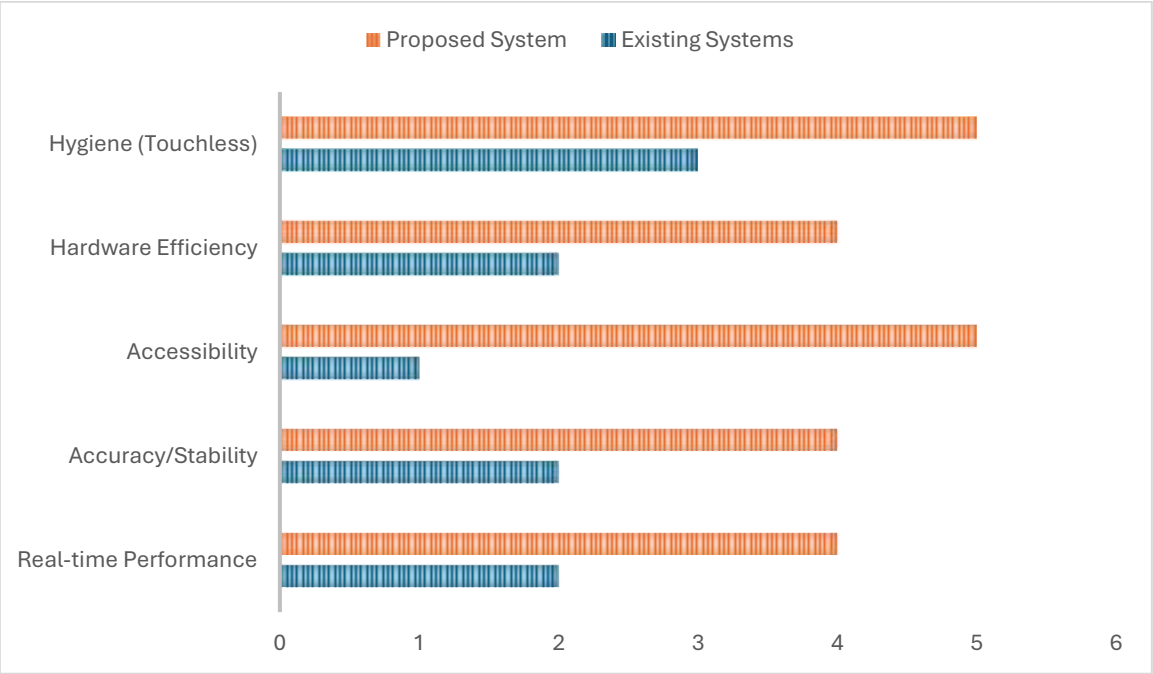
The proposed system was compared with existing touchless interaction systems based on key parameters such as real-time performance, stability, accessibility, hardware requirements, and hygienic interaction. The comparison results, as presented in Table 7.2 and Figure 7.1, indicate that most existing systems are limited to single-modality interaction and lack control stabilization mechanisms.

Table 6 Analysis of Existing and Proposed Systems

Feature	Existing Systems	Proposed System
Interaction Type	Single modality (face or hand)	Hybrid (face + hand)
Real-time Performance	Moderate	High
Hardware Requirement	May require GPU	Works on standard CPU
Control Stability	Limited	Improved (clutch/pause mechanism)
Accessibility	Limited focus	Designed for motor-impaired users
Hygiene	Partial	Fully touchless

In contrast, the proposed system introduces a hybrid interaction approach by integrating both face and hand gesture control, resulting in improved flexibility and reliability. The system operates efficiently on standard hardware without requiring specialized resources, making it suitable for edge-based deployment.

The graphical comparison further highlights that the proposed system outperforms existing solutions in terms of accessibility, stability, and hygienic interaction. Overall, the proposed approach provides a more comprehensive and practical solution compared to traditional touchless systems.



Graph 1 Comparison Graph

7.4 Discussion

The results demonstrate that the proposed system effectively addresses the limitations identified in existing solutions. By integrating both face and hand gesture interaction, the system provides greater flexibility compared to single-modality approaches.

The use of lightweight computer vision techniques ensures real-time performance without requiring high-end hardware, making the system suitable for practical deployment. The addition of control stabilization mechanisms enhances usability, particularly for users with limited motor control.

However, the system still faces challenges in extreme lighting conditions and may require further optimization for improved robustness. Despite these limitations, the proposed system provides a significant improvement over existing approaches in terms of accessibility, usability, and hygienic interaction.

Chapter 8

Conclusion and Future Work

This chapter summarizes the overall outcomes of the proposed system and highlights the key contributions achieved. It also discusses potential improvements and future enhancements that can further improve the system's performance, scalability, and applicability.

8.1 Conclusion

The proposed computer vision-based touchless device control system successfully demonstrates an alternative approach to traditional human–computer interaction methods. By integrating face-based and hand gesture-based interaction into a unified hybrid framework, the system enables users to control devices without physical contact.

The system effectively achieves real-time performance using lightweight computer vision techniques and operates efficiently on standard hardware without requiring specialized resources. Features such as head movement-based cursor control, blink-based clicking, hand gesture recognition, and the implementation of clutch and pause mechanisms contribute to improved usability and interaction stability.

The proposed solution addresses key challenges identified in existing systems, including lack of hybrid interaction, limited accessibility, and absence of control stabilization. It also supports

hygienic interaction in shared environments and enhances accessibility for motor-impaired users. Overall, the system provides a practical, efficient, and user-friendly touchless interaction solution.

8.2 Future Work

Several improvements can be considered to further improve the current system's capabilities.

- Improving robustness under varying lighting conditions using adaptive thresholding or advanced vision techniques.
- Incorporating machine learning-based gesture classification for more dynamic and complex gesture recognition.
- Expanding gesture vocabulary to support additional system functionalities and applications.
- Enhancing accuracy and stability through smoothing algorithms and noise reduction techniques.
- Developing a user-specific calibration system to adapt sensitivity based on individual preferences.
- Deploying the system on embedded or portable devices for use in kiosks and public environments.
- Integrating voice-based interaction to extend the system into a fully multimodal interface.

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